Experiment 6: Interfacing Sensors and Actuators with Arduino Uno – Door Opener Using Ultrasonic Sensor and Servo Motor

```
#include <Servo.h>

const int trigPin = 9;
const int echoPin = 8;
const int servoPin = 6;

long duration;
int distance;

Servo myServo;

void setup()
{
   pinMode(trigPin, OUTPUT);
   pinMode(echoPin, INPUT);
   myServo.attach(servoPin);
   Serial.begin(9600);
   myServo.write(0);
   delay(500);
}

void loop()
{
   digitalWrite(trigPin, LOW);
   delayMicroseconds(2);

   digitalWrite(trigPin, HIGH);
   delayMicroseconds(10);

   digitalWrite(trigPin, LOW);

   duration = pulseIn(echoPin, HIGH);
   distance = duration * 0.0343 / 2;

   Serial.print("Distance: ");
   Serial.print(distance);
   Serial.println(" cm");

   if (distance < 20 && distance > 0)
   {
      Serial.println("Object detected! Opening door...");
      myServo.write(90);
```

```
    }
    else
    {
      Serial.println("Path is clear. Closing door.");
      myServo.write(0);
    }

    delay(100);
}
```

